

Install-count telemetry — under-counting (to revisit)

Status: **DIAGNOSED**, fix not yet implemented · Opened 2026-07-15

Symptom

~225 installs reported, but the portal install funnel shows 1.

Verdict: the portal is NOT the bug

The portal frontend + backend logic are **correct**. The count faithfully reports the number of distinct `install_id`s that registered:

- `POST /api/install/register` (`portal/app/installs.py`) is an **idempotent upsert keyed on `install_id`** (the primary key) — one row per distinct install, by design.
- The admin count (`GET /api/admin/installs`) is a straight `count(install_id)` per state — no over-collapsing `DISTINCT`, no hidden filter.
- The frontend just renders the returned number.

So "225 → 1" means **only one distinct `install_id` ever reached the portal**. The problem is entirely **upstream** — in how installs generate/report their id.

Root causes (upstream)

#1 — winget/MSI installs never phone home

- `scoop / brew / deb / rpm` run `packaging/common/phone-home.{ps1,sh}` on install → they register a `DOWNLOADED` event with a **unique per-machine id**.
- The **Windows MSI has no phone-home** — there is **no custom action** in `packaging/windows/cloudlearn.wxs`. So a winget/MSI install is **invisible** to the portal until the user actually runs `vyomi up` and the appliance registers itself.
- Net: N downloads/installs but few boots → the portal only sees the boots.

#2 — Windows/Max path: the appliance `install_id` can collapse

- The appliance id comes from `core/license_remote.get_or_create_install_id(state)` :
 1. `STATE['install_id']` if already set,
 2. `$VYOMI_INSTALL_ID` env (forwarded from the package-manager id),
 3. fallback `SHA-256(hostname | /data-inode | uuid.getnode()-MAC)[:24]`.
- The **bash launcher** (`scripts/cloud-learn`) forwards `$VYOMI_INSTALL_ID` from `~/vyomi/install_id`, **but the Windows PowerShell launcher AND the new Go launcher (`cmd/vyomi/`) do NOT forward it** into the VM/appliance container. (The Windows path is `vyomi.exe` → `multipass exec ... bash cloud-learn up`, and the host's id is never synced into the VM.)

- So on Windows/Max the appliance falls to the **hardware-hash fallback**, whose inputs (container hostname / a fresh `/data` volume inode / docker-assigned MAC) can be **identical across cloned/standardized appliance containers** → multiple real machines share one `install_id` → they merge into a single portal row.

Proposed fix (ships in the same `vyomi.exe` / MSI)

1. **Make `vyomi.exe` phone home itself** — compiled (winget-safe, no script): generate + persist a unique id at first run and `POST /api/install/register` (state `DOWNLOADED` / `INSTALLED`). Closes gap #1 so MSI/winget installs count.
2. **Forward the id into the appliance** — pass `VYOMI_INSTALL_ID` through the `multipass exec ... cloud-learn up` invocation (and the PS1 launcher, for parity with the bash one) so `DOWNLOADED`→`INSTALLED` link and each machine stays distinct.
3. **Harden the fallback** — in `get_or_create_install_id`, replace the collapsing hardware hash with a **persisted random UUID** (stable across restarts via the `/data` state, unique per install) so distinct installs can never merge.

To do when we revisit

- ☐ Implement fixes 1–3 above.
- ☐ Reset the portal `applianceinstall` table for a clean baseline (backup first — `CREATE TABLE applianceinstall_backup_<date> AS TABLE applianceinstall;` then `DELETE FROM applianceinstall;` via the Render `cloudlearn-data` DB console).
- ☐ Re-observe: each real install (MSI/winget included) should now produce a distinct row; downloads-that-never-boot become visible as `DOWNLOADED`.

Evidence / file refs

- Portal register + count: `portal/app/installs.py` (`register_install`, `admin_installs`).
- Portal model (PK `install_id`): `portal/app/models.py` (`ApplianceInstall`).
- Appliance id: `appliance/core/license_remote.py:482` (`get_or_create_install_id`).
- Phone-home client (has unique id; not run by MSI): `packaging/common/phone-home.ps1`.
- MSI (no phone-home custom action): `packaging/windows/cloudlearn.wxs`.
- Launcher id-forwarding present in bash only: `scripts/cloud-learn` (~L142–196); absent in `scripts/cloud-learn.ps1` and `cmd/vyomi/`.